

applications in other domains too.

- **Agriculture:** Embraces IoT for precision farming, improving crop yields and sustainability
- **Smart cities:** Leverage IoT for traffic management, waste optimisation, and environmental monitoring, making urban areas more efficient and sustainable
- **Retail:** Uses IoT for inventory management and personalised shopping experiences
- **Energy management:** This is enhanced by IoT, with smart grids and energy consumption monitoring promoting sustainability
- **Environmental monitoring:** IoT sensors aid in tracking wildlife, water quality, and pollution levels, contributing to conservation efforts
- **Smart buildings:** Utilise IoT for energy optimisation and security
- **Supply chain management:** Relies on real-time tracking and inventory optimisation
- **Security and surveillance:** IoT enhances security and surveillance with advanced camera and sensor technologies

IoT also plays a crucial role in water management, tourism, education, wearables, and sports and entertainment, offering a wide range of benefits across these domains.

Key protocols in IoT implementations

In the world of IoT implementations, protocols play a pivotal role in enabling devices to communicate, exchange data, and function seamlessly within the IoT ecosystem. One of the foundational protocols is MQTT (Message Queuing Telemetry Transport), known for its lightweight and efficient nature. MQTT is ideal for scenarios where low bandwidth and minimal power consumption are crucial. This makes it a popular choice for IoT devices, especially in remote and resource-constrained environments.



Figure 1: Use cases and applications of Internet of Things

HTTP (Hypertext Transfer Protocol), a protocol well-known in web communication, also plays a significant role in IoT. With the emergence of RESTful APIs, IoT devices can communicate with web servers over HTTP, enabling seamless integration with web applications and services. This simplifies data exchange and provides a familiar interface for developers.

On the other hand, CoAP (Constrained Application Protocol) is designed explicitly for IoT devices with limited resources. It operates like HTTP but with optimisations for low-power and constrained environments. CoAP is well-suited for IoT implementations where efficiency and scalability are paramount.

For IoT applications that require real-time communication and low latency, WebSocket is a critical protocol. It enables full-duplex communication between devices and servers, allowing for instantaneous data exchange. This protocol is particularly valuable in scenarios like remote control and monitoring.

AMQP (Advanced Message Queuing Protocol) is another protocol that finds its place in IoT implementations, especially in scenarios requiring reliable message queuing and routing. It provides a robust framework for message-oriented middleware, ensuring messages reach their destinations efficiently and safely.

In industrial and enterprise IoT applications, Modbus and BACnet are prevalent protocols. Modbus is used extensively in industrial control systems and automation, facilitating communication between devices like sensors and programmable logic controllers. BACnet, on the other hand, is focused on building automation and HVAC systems, ensuring interoperability and efficient data exchange in smart buildings.



Figure 2: Key protocols and implementation scenarios for IoT